

Project Design Phase Proposed Solution Template

Date	10 July 2026
Team ID	
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers often struggle to identify the most suitable crop due to variations in soil nutrients and environmental conditions. Traditional decision-making methods may lead to low crop yield, inefficient resource utilization, and financial losses.
2.	Idea / Solution description	Develop OptiCrop , an AI-powered crop recommendation system that uses Machine Learning to analyze Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall. The system recommends the most suitable crop through a user-friendly Flask web application.
3.	Novelty / Uniqueness	Combines multiple machine learning algorithms to identify the best-performing model for crop prediction. Provides fast, accurate, and data-driven recommendations through a simple web interface that is easy for farmers to use.
4.	Social Impact / Customer Satisfaction	Helps farmers make informed farming decisions, increases agricultural productivity, reduces crop failure risks, optimizes the use of soil nutrients and water resources, and promotes sustainable agriculture.
5.	Business Model (Revenue Model)	Freemium model for farmers with premium analytics and advisory services. Subscription plans for agribusiness companies and agricultural organizations. Future revenue through government partnerships, API integration, and agricultural consultancy services.
6.	Scalability of the Solution	The system can be extended with fertilizer recommendation, plant disease detection, weather forecasting, IoT sensor integration, multilingual support, mobile applications, and cloud deployment for large-scale agricultural use.